

Notes: Heckman (Econometrica, 1979)

Sample Selection Bias as a Specification Error

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1 Big picture

- **Question.** What bias arises when researchers estimate behavioral relationships using nonrandomly selected samples, and how can that bias be corrected with simple regression tools?
- **Core insight.** Sample selection bias can be written as an ordinary omitted-variable problem. The selected-sample regression omits a conditional expectation term that depends on the probability of selection.
- **Main contribution.** The paper develops a simple consistent two-step estimator for censored samples: estimate the selection equation by probit, compute the inverse Mills ratio, and include it as an extra regressor in the outcome equation.
- **Why it matters.** Many observed outcomes are only observed after a participation decision: wages only for workers, union wages only for union members, migrant wages only for migrants, trainee earnings only for trainees, and panel outcomes only for nonattriters.
- **Problem with naive OLS.** OLS on the observed subsample estimates the structural equation plus a selection term. Unless that term is constant or independent of regressors, slope coefficients are biased.
- **Econometric mechanism.** If outcome unobservables and selection unobservables are correlated, conditioning on selection shifts the mean of the outcome error. That conditional mean is the omitted variable.
- **Normality result.** Under joint normality, the omitted conditional mean is proportional to the inverse Mills ratio.
- **Two-step estimator.** Estimate selection by probit in the full sample; compute the inverse Mills ratio for selected observations; regress the observed outcome on original regressors plus the Mills ratio.
- **Testing selection.** The coefficient on the Mills ratio is proportional to the covariance between outcome and selection errors. If it is zero, there is no selection bias under the model.
- **Important warning.** Because the Mills ratio is estimated in the first stage, standard OLS standard errors in the second stage are generally wrong unless the null of no selection bias holds.
- **Takeaway.** Selection bias is not a mysterious property of nonrandom samples. It is a specification error caused by omitting a selection correction term.

2 Data structure and model primitives

2.1 Why sample selection creates a missing-data problem

- The researcher wants to estimate a behavioral equation for an outcome Y_{1i} .
- But Y_{1i} is observed only for a selected subset of observations.
- Selection can arise from individual behavior, such as labor force participation, migration, union membership, or training take-up.
- Selection can also arise from the research design, such as dropping attriters, retaining only intact families in panels, or conditioning on survival.
- The crucial question is why observations are missing. Random missingness loses efficiency; nonrandom missingness changes the conditional mean of the regression disturbance.

Interpretation: The paper's point is that the selection rule determines whether the observed subsample has the same regression function as the population. If not, OLS estimates the wrong conditional expectation.

2.2 Two-equation setup

The paper starts with two latent structural equations:

$$Y_{1i} = X_{1i}\beta_1 + U_{1i}, \quad (1)$$

$$Y_{2i} = X_{2i}\beta_2 + U_{2i}. \quad (2)$$

- Y_{1i} is the outcome equation of interest.
- Y_{2i} is the selection or participation index.
- X_{1i} and X_{2i} are exogenous variables.
- The researcher observes Y_{1i} only when $Y_{2i} \geq 0$.
- The threshold zero is a normalization.

Interpretation: This structure is general enough to cover many settings. For example, Y_{1i} can be market wage and Y_{2i} can be the latent labor-force participation index.

2.3 Population regression versus selected-sample regression

If the full population were observed, the regression function would be

$$E(Y_{1i} \mid X_{1i}) = X_{1i}\beta_1. \quad (3)$$

But in the selected sample, the regression function is

$$E(Y_{1i} \mid X_{1i}, Y_{2i} \geq 0) = X_{1i}\beta_1 + E(U_{1i} \mid U_{2i} \geq -X_{2i}\beta_2). \quad (4)$$

Interpretation: The second term is the selection correction. If the researcher omits it, the second-stage regression has an omitted-variable problem.

2.4 When OLS is valid

Naive OLS on the selected sample is valid only under special cases:

- **Random selection:** U_{1i} is independent of the selection event, so $E(U_{1i} \mid Y_{2i} \geq 0) = 0$.
- **Only intercept bias:** selection probability is the same for everyone, so the conditional mean is constant and only the intercept is affected.
- **No correlation in errors:** the outcome error and selection error are uncorrelated under the maintained distributional model.

Interpretation: The dangerous case is when selection varies with observables and is correlated with unobservables in the outcome equation. Then slope coefficients are biased, not just the intercept.

3 Selection bias as omitted-variable bias

3.1 Selected-sample conditional mean

The selected-sample regression function is

$$E(Y_{1i} \mid X_{1i}, Y_{2i} \geq 0) = X_{1i}\beta_1 + E(U_{1i} \mid U_{2i} \geq -X_{2i}\beta_2). \quad (5)$$

- The second term depends on the selection equation.

- It is generally a nonlinear function of $X_{2i}\beta_2$.
- If variables in X_{2i} are omitted from X_{1i} , those variables may appear to affect Y_{1i} through selection rather than through the structural outcome equation.

Interpretation: This is why selection bias can make irrelevant variables look significant in a selected-sample regression.

3.2 Examples from the paper

- **Working women:** observed market wages are wages of women whose market wage exceeds the home wage at zero hours of work.
- **Union members:** observed union wages are wages for workers who found union membership preferable to nonunion alternatives.
- **Migrants:** migrant wages do not directly reveal what nonmigrants would earn if they migrated.
- **Training programs:** trainee earnings do not directly reveal what nontrainees would earn if they enrolled.
- **Panel attrition:** using only observations followed for the full sample can select on unobservable persistence or outcomes.

Interpretation: In each case, the observed outcome sample is not a random draw from the population relevant for the behavioral equation.

4 Normal model and the inverse Mills correction

4.1 Joint normality

Assume that (U_{1i}, U_{2i}) is bivariate normal with covariance σ_{12} and variance σ_{22} for U_{2i} . Let

$$Z_i = \frac{X_{2i}\beta_2}{\sqrt{\sigma_{22}}}. \quad (6)$$

The inverse Mills ratio is

$$\lambda_i = \frac{\phi(Z_i)}{\Phi(Z_i)}, \quad (7)$$

where ϕ and Φ are the standard normal density and distribution functions.

Interpretation: Heckman's notation writes the denominator as $1 - \Phi(-Z_i)$, which is the same as $\Phi(Z_i)$.

4.2 Conditional mean of the omitted variable

Under normality,

$$E(U_{1i} \mid U_{2i} \geq -X_{2i}\beta_2) = \frac{\sigma_{12}}{\sqrt{\sigma_{22}}} \lambda_i. \quad (8)$$

Therefore the selected-sample outcome equation becomes

$$Y_{1i} = X_{1i}\beta_1 + C\lambda_i + V_{1i}, \quad C = \frac{\sigma_{12}}{\sqrt{\sigma_{22}}}. \quad (9)$$

Interpretation: The coefficient on λ_i measures the covariance between the outcome error and the selection error. If $C \neq 0$, selection matters.

4.3 Why the inverse Mills ratio is an omitted variable

- Selection depends on $X_{2i}\beta_2 + U_{2i}$.
- Conditional on being selected, U_{2i} has a shifted mean.
- If U_{1i} is correlated with U_{2i} , then U_{1i} also has a shifted mean.
- Under normality, that shifted mean is linear in the inverse Mills ratio.

Interpretation: The inverse Mills ratio is the observable proxy for the conditional mean of the unobserved outcome error induced by selection.

5 The two-step Heckman estimator

5.1 Step 1: Estimate the selection equation

Define the selection indicator

$$D_i = 1\{Y_{2i} \geq 0\}. \quad (10)$$

Estimate the probit model

$$P(D_i = 1 \mid X_{2i}) = \Phi(X_{2i}\gamma), \quad \gamma = \frac{\beta_2}{\sqrt{\sigma_{22}}}. \quad (11)$$

Then compute

$$\hat{\lambda}_i = \frac{\phi(X_{2i}\hat{\gamma})}{\Phi(X_{2i}\hat{\gamma})}. \quad (12)$$

Interpretation: The first step recovers the selection probability and therefore the Mills-ratio correction for each selected observation.

5.2 Step 2: Estimate the outcome equation

For observations with $D_i = 1$, run

$$Y_{1i} = X_{1i}\beta_1 + C\hat{\lambda}_i + \text{error}_i. \quad (13)$$

- The coefficient on X_{1i} estimates the structural outcome parameters.
- The coefficient on $\hat{\lambda}_i$ estimates $C = \sigma_{12}/\sqrt{\sigma_{22}}$.
- A nonzero C indicates selection bias under the model.

Interpretation: This is the classic Heckman selection correction. It lets the researcher use linear regression after estimating one nonlinear selection equation.

5.3 Estimating the error variance

The paper also gives a consistent estimator for the population variance σ_{11} . The key adjustment corrects the selected-sample residual variance for the variance reduction caused by truncation and for the estimated selection coefficient.

Interpretation: The raw second-stage residual variance is not enough because the selected-sample disturbance is heteroskedastic and has lower variance than the population disturbance.

5.4 Generated regressor and standard errors

The second-stage Mills ratio is estimated, not observed. Therefore ordinary OLS standard errors are generally incorrect. The paper derives the limiting distribution of the estimator and shows how to estimate the correct asymptotic covariance matrix.

- If $C = 0$, the usual regression standard errors are appropriate.
- If $C \neq 0$, the usual second-stage OLS standard errors understate true standard errors.
- The correction requires accounting for first-stage probit estimation error.

Interpretation: The two-step estimator is easy to compute, but valid inference requires the Heckman standard-error correction unless one is exactly testing the null of no selection.

6 Identification and empirical design

6.1 Sources of identification

- **Functional-form identification:** under joint normality, nonlinearity in λ_i identifies the correction term even if X_1 and X_2 overlap.
- **Exclusion restrictions:** stronger identification comes from variables in X_2 that affect selection but do not directly affect the outcome.
- **Support:** variation in selection probabilities helps distinguish the Mills ratio from the outcome regressors.

Interpretation: Modern applications usually prefer exclusion restrictions because relying only on the normality-induced nonlinearity can be fragile.

6.2 Censored versus truncated samples

- In a **censored sample**, the researcher observes whether an observation is selected and observes X_2 for both selected and unselected observations.
- In a **truncated sample**, unselected observations may be missing entirely.
- Heckman's two-step estimator in this paper is designed for the censored sample case, where the first-stage probit can be estimated on the full sample.

Interpretation: The distinction matters because the first-stage selection model requires data on selected and unselected observations.

6.3 Testing for selection bias

A common diagnostic is the estimated coefficient on the inverse Mills ratio:

$$H_0 : C = 0. \tag{14}$$

- If $C = 0$, selection and outcome errors are uncorrelated.
- Under the normal model, selected-sample OLS is not biased by selection.
- If $C \neq 0$, omitting λ_i produces specification error.

Interpretation: The Mills-ratio coefficient is both a correction term and a test statistic for sample selection.

6.4 What the method assumes

- The selection rule is correctly specified.
- The first-stage probit model is correctly specified.
- The joint normality assumption provides the inverse Mills ratio form.
- Observations are randomly sampled before selection occurs.
- Exclusion restrictions, when used, are valid.

Interpretation: The method is powerful, but it moves the burden to correct modeling of the selection mechanism.

6.5 Limitations and modern interpretation

- The method can be sensitive to distributional assumptions when no strong exclusion restriction is available.
- It estimates a parametric correction, not a fully nonparametric selection process.
- It corrects mean selection under the model, but it does not automatically solve treatment-effect heterogeneity or causal identification problems.
- It is most useful when there is a clear selection equation and variables excluded from the outcome equation.

Interpretation: The paper is foundational because it reframes sample selection as a specification problem and gives researchers a tractable correction. Later work generalizes and relaxes parts of the framework.

7 Original equation panels

7.1 Panel 1: Problem definition and selection model

This paper discusses the bias that results from using nonrandomly selected samples to estimate behavioral relationships as an ordinary specification error or “omitted variables” bias. A simple consistent two stage estimator is considered that enables analysts to utilize simple regression methods to estimate behavioral functions by least squares methods. The asymptotic distribution of the estimator is derived.

(a) Original abstract statement of specification-error interpretation.

Consider a random sample of I observations. Equations for individual i are

$$(1a) \quad Y_{1i} = X_{1i}\beta_1 + U_{1i},$$

$$(1b) \quad Y_{2i} = X_{2i}\beta_2 + U_{2i} \quad (i = 1, \dots, I),$$

where X_{ji} is a $1 \times K_j$ vector of exogenous regressors, β_j is a $K_j \times 1$ vector of parameters, and

$$E(U_{ji}) = 0, \quad E(U_{ji}U_{j'i'}) = \sigma_{ji'}, \quad i = i',$$

(b) Original two-equation model.

Read:

- The abstract states the key idea: nonrandom selection produces an omitted-variable style specification error.
- The two-equation model separates the outcome equation from the selection equation.

Economic message: Selection bias is not merely sample-size loss. It changes the regression function unless selection is random with respect to the outcome error.

7.2 Panel 2: Selected-sample regression function

$$\begin{aligned} E(U_{1i} | X_{1i}, \text{sample selection rule}) &= E(U_{1i} | X_{1i}, Y_{2i} \geq 0) \\ &= E(U_{1i} | X_{1i}, U_{2i} \geq -X_{2i}\beta_2). \end{aligned}$$

case of independence between U_{1i} and U_{2i} , so that the data on Y_{1i} is randomly, the conditional mean of U_{1i} is zero. In the general case and the subsample regression function is

$$E(Y_{1i} | X_{1i}, Y_{2i} \geq 0) = X_{1i}\beta_1 + E(U_{1i} | U_{2i} \geq -X_{2i}\beta_2).$$

Figure 2: Original selected-sample conditional mean.

Read:

- The selected-sample regression has the structural term $X_{1i}\beta_1$ plus a conditional mean term.
- The conditional mean depends on the selection rule.

Economic message: The omitted conditional expectation is the source of selection bias.

7.3 Panel 3: Normal model and inverse Mills ratio

$$\begin{aligned} E(Y_{1i} | X_{1i}, Y_{2i} \geq 0) &= X_{1i}\beta_1 + \frac{\sigma_{12}}{(\sigma_{22})^{\frac{1}{2}}} \lambda_i, \\ E(Y_{2i} | X_{2i}, Y_{2i} \geq 0) &= X_{2i}\beta_2 + \frac{\sigma_{22}}{(\sigma_{22})^{\frac{1}{2}}} \lambda_i, \\ E(U_{1i} | U_{2i} \geq -X_{2i}\beta_2) &= \frac{\sigma_{12}}{(\sigma_{22})^{\frac{1}{2}}} \lambda_i, \\ E(U_{2i} | U_{2i} \geq -X_{2i}\beta_2) &= \frac{\sigma_{22}}{(\sigma_{22})^{\frac{1}{2}}} \lambda_i, \\ \lambda_i &= \frac{\phi(Z_i)}{1 - \Phi(Z_i)} = \frac{\phi(Z_i)}{\Phi(-Z_i)}, \\ E(V_{1i} | X_{1i}, \lambda_i, U_{2i} \geq -X_{2i}\beta_2) &= 0, \\ E(V_{2i} | X_{2i}, \lambda_i, U_{2i} \geq -X_{2i}\beta_2) &= 0, \\ E(V_{ji} V_{j'i'} | X_{1i}, X_{2i}, \lambda_i, U_{2i} \geq -X_{2i}\beta_2) &= 0, \end{aligned}$$

(a) Original inverse Mills ratio definition. (b) Original conditional regression equations.

Read:

- Joint normality turns the omitted conditional mean into a coefficient times the inverse Mills ratio.
- The selected outcome equation can be estimated by adding λ_i as a regressor.

Economic message: This is the mathematical bridge from selection bias to a practical correction.

7.4 Panel 4: Heteroskedasticity, variance bias, and the two-step correction

general case requires some argument. Note that equation (4a) with an estimated value of λ_i used in place of the true value of λ_i may be written as

$$(4f) \quad E(V_{1i}^2 | X_{1i}, \lambda_i, U_{2i} \geq -X_{2i}\beta_2) = \sigma_{11}((1-\rho^2) + \rho^2(1 + Z_i\lambda_i - \lambda_i^2)),$$

$$(4g) \quad E(V_{1i}V_{2i} | X_{1i}, X_{2i}, \lambda_i, U_{2i} \geq -X_{2i}\beta_2) = \sigma_{12}(1 + Z_i\lambda_i - \lambda_i^2),$$

$$(4h) \quad E(V_{2i}^2 | X_{2i}, \lambda_i, U_{2i} \geq -X_{2i}\beta_2) = \sigma_{22}(1 + Z_i\lambda_i - \lambda_i^2),$$

where

$$\rho^2 = \frac{\sigma_{12}^2}{\sigma_{11}\sigma_{22}}$$

and

$$(5) \quad 0 \leq 1 + \lambda_i Z_i - \lambda_i^2 \leq 1.$$

(a) Original variance and covariance expressions.

In practice, one does not know λ_i . But in the case of a censored sample, in which one does not have information on Y_{1i} if $Y_{2i} \leq 0$, but one does know X_{2i} for observations with $Y_{2i} \leq 0$, one can estimate λ_i by the following procedure:

(1) Estimate the parameters of the probability that $Y_{2i} \geq 0$ (i.e., $\beta_2/(\sigma_{22})^{1/2}$) using probit analysis for the full sample.³

(2) From this estimator of $\beta_2/(\sigma_{22})^{1/2}$ ($= \beta_2^*$) one can estimate Z_i and hence λ_i . All of these estimators are consistent.

(3) The estimated value of λ_i may be used as a regressor in equation (4a) fit on the selected subsample. Regression estimators of equation (4a) are *consistent* for β_1 and $\sigma_{12}/(\sigma_{22})^{1/2}$ (the coefficients of X_{1i} and λ_i , respectively).⁴

(4) One can consistently estimate σ_{11} by the following procedure. From step 3, one consistently estimates $C = \rho(\sigma_{11})^{1/2} = \sigma_{12}/(\sigma_{22})^{1/2}$. Denote the residual for the i th observation obtained from step 3 as $\hat{V}_{1i}$ and the estimator of C by $\hat{C}$. Then an estimator of σ_{11} is

$$\hat{\sigma}_{11} = \frac{\sum_{i=1}^{I_1} \hat{V}_{1i}^2}{I_1} - \frac{\hat{C}}{I_1} \sum_{i=1}^{I_1} (\hat{\lambda}_i \hat{Z}_i - \hat{\lambda}_i^2)$$

(b) Original two-step variance-correction discussion.

Read:

- The selected-sample error is heteroskedastic.
- The raw OLS residual variance is downward biased for the population variance.
- The paper supplies a consistent variance estimator.

Economic message: Correcting the conditional mean is not enough for inference. The variance and standard errors also need adjustment.

7.5 Panel 5: Asymptotic distribution and standard-error correction

(a) Limit distribution of $\hat{\beta}_1$

$$\text{plim}_{I \rightarrow \infty} I_1 \begin{pmatrix} \Sigma X'_{1i} X_{1i} & \Sigma X'_{1i} \hat{\lambda}_i \\ \Sigma X_{1i} \hat{\lambda}_i & \Sigma \hat{\lambda}_i^2 \end{pmatrix}^{-1} = \text{plim}_{I \rightarrow \infty} I_1 \begin{pmatrix} \Sigma X'_{1i} X_{1i} & \Sigma X'_{1i} \lambda_i \\ \Sigma X_{1i} \lambda_i & \Sigma \lambda_i^2 \end{pmatrix}^{-1} = B,$$

where B is a finite positive definite matrix.⁷ Under these assumptions,

$$\sqrt{I_1} (\hat{\beta}_1 - \beta_1) \sim N(0, B\psi B')$$

where

$$\psi = \text{plim}_{I \rightarrow \infty} \left[\sigma_{11} \begin{pmatrix} \frac{\Sigma X'_{1i} X_{1i} \eta_i}{I_1} & \frac{\Sigma X'_{1i} \lambda_i \eta_i}{I_1} \\ \frac{\Sigma \lambda_i X_{1i} \eta_i}{I_1} & \frac{\Sigma \lambda_i^2 \eta_i}{I_1} \end{pmatrix} + C^2 \left(\frac{I_1}{I} \right) \begin{pmatrix} \sum_{i=1, i' \neq 1}^{I_1} \frac{X'_{1i} X_{1i'} \theta_{ii'}}{I_1^2} & \sum_{i=1, i' \neq 1}^{I_1} \frac{X'_{1i} \pi_{ii'}}{I_1^2} \\ \sum_{i'=1, i \neq 1}^{I_1} \frac{X_{1i'} \pi_{ii'}}{I_1^2} & \sum_{i=1, i' \neq 1}^{I_1} \frac{\Omega_{ii'}}{I_1^2} \end{pmatrix} \right]$$

$$\text{plim}_{I \rightarrow \infty} \frac{I_1}{I} = k, \quad 0 < k < 1,$$

where

$$C = \sigma_{12}/(\sigma_{22})^{1/2},$$

$$\eta_i = (1 + C^2(Z_i\lambda_i - \lambda_i^2)/\sigma_{11}),$$

$$\pi_{ii'} = \left(\frac{\partial \lambda_i}{\partial Z_i} \right) \left(\frac{\partial \lambda_{i'}}{\partial Z_{i'}} \right) \lambda_i X_{2i} \Sigma X'_{2i'},$$

$$\theta_{ii'} = \left(\frac{\partial \lambda_i}{\partial Z_i} \right) \left(\frac{\partial \lambda_{i'}}{\partial Z_{i'}} \right) X_{2i} \Sigma X'_{2i'},$$

$$\Omega_{ii'} = (\lambda_i \lambda_{i'}) \left(\frac{\partial \lambda_i}{\partial Z_i} \right) \left(\frac{\partial \lambda_{i'}}{\partial Z_{i'}} \right) X_{2i} \Sigma X'_{2i'},$$

where $\partial \lambda_i / \partial Z_i$ is the derivative of λ_i with respect to Z_i ,

general case requires some argument. Note that equation (4a) with an estimated value of λ_i used in place of the true value of λ_i may be written as

$$(4a') \quad Y_{1i} = X_{1i}\beta_1 + C\hat{\lambda}_i + C(\lambda_i - \hat{\lambda}_i) + V_{1i}.$$

The error term in the equation consists of the final two terms in the equation.

Since λ_i is estimated by $\beta_2/(\sigma_{22})^{1/2}$ ($= \beta_2^*$) which is estimated from the entire sample of I observations by a maximum likelihood probit analysis,⁶ and since λ_i is a twice continuously differentiable function of β_2^* , $\sqrt{I}(\lambda_i - \hat{\lambda}_i)$ has a well defined limiting normal distribution

$$\sqrt{I}(\lambda_i - \hat{\lambda}_i) \sim N(0, \Sigma_i)$$

where Σ_i is the asymptotic variance-covariance matrix obtained from that of β_2^* by the following equation:

$$\Sigma_i = \left(\frac{\partial \lambda_i}{\partial \beta_2} \right)^2 X_{2i} \Sigma X'_{2i},$$

where $\partial \lambda_i / \partial \beta_2$ is the derivative of λ_i with respect to β_2 , and Σ is the asymptotic variance-covariance matrix of $\sqrt{I}(\beta_2^* - \beta_2)$.

We seek the limiting distribution of

$$\sqrt{I_1} \left(\frac{\hat{\beta}_1 - \beta_1}{\hat{C} - C} \right) = I_1 \begin{pmatrix} \Sigma X'_{1i} X_{1i} & \Sigma X'_{1i} \hat{\lambda}_i \\ \Sigma X_{1i} \hat{\lambda}_i & \Sigma \hat{\lambda}_i^2 \end{pmatrix}^{-1} \frac{1}{\sqrt{I_1}} \begin{pmatrix} \Sigma X'_{1i} (C(\lambda_i - \hat{\lambda}_i) + V_{1i}) \\ \Sigma \lambda_i (C(\lambda_i - \hat{\lambda}_i) + V_{1i}) \end{pmatrix}.$$

(a) Generated-regressor component.

(b) Asymptotic variance matrix.

Read:

- The first-stage probit makes the inverse Mills ratio an estimated regressor.
- The paper derives the asymptotic variance of the second-stage coefficients.
- If the Mills coefficient is nonzero, naive OLS standard errors understate true uncertainty.

Economic message: The two-step estimator is computationally simple, but its inference is not the same as ordinary OLS inference.

8 Short summary

- The paper treats sample selection bias as a specification error.
- The selected-sample regression function differs from the population regression function because the outcome error has a nonzero conditional mean after selection.
- Under joint normality, the conditional mean is proportional to the inverse Mills ratio.
- The two-step estimator first estimates selection by probit, then includes the estimated Mills ratio in the outcome equation.
- The coefficient on the Mills ratio tests whether selection and outcome errors are correlated.
- Standard second-stage OLS standard errors are generally incorrect because the Mills ratio is generated in the first stage.
- The estimator is simple, consistent, and useful for censored samples where selection status and selection covariates are observed for everyone.
- The method provides a foundation for empirical work on labor force participation, migration, treatment programs, attrition, and many other settings with nonrandom observation of outcomes.

9 Conclusion

9.1 Core conclusion

Heckman shows that sample selection bias can be understood as an omitted-variable problem. When the sample is selected on a latent index whose unobservable component is correlated with the outcome error, the selected-sample regression omits the conditional mean of the outcome disturbance.

9.2 Economic intuition

The observed sample is not representative of the population. Workers, migrants, trainees, union members, and nonattriters are selected in ways that depend on observed and unobserved factors. If the unobserved factors in the selection decision are correlated with unobserved determinants of the outcome, the outcome regression error is no longer mean zero in the observed sample.

9.3 Takeaway

The paper's lasting contribution is the two-step correction: estimate the selection probability, compute the inverse Mills ratio, and include it in the outcome regression. The method is simple enough for applied work, but the paper also emphasizes that valid inference requires accounting for the generated selection correction.